

Lee Luderman

Software Engineer | Data Scientist | Expertise in Python, R, and MATLAB

About Me

I am a versatile professional with a strong background in data analysis, software development, and financial management. My recent roles involved developing automation solutions with Python and creating a responsive website using Next.js and React. With a master's degree in mathematics and expertise in MATLAB, R, and Python, I excel at transforming complex data into actionable insights. I am currently seeking software engineering, data science, and data analytics opportunities in Sydney.

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🚗 Car Available, Driving License C

Programming Languages

🐍 Python

📊 MATLAB

📊 R

📦 Git

⚙️ React

📄 HTML

🎨 CSS

📌 C/C++

EDUCATION

Completed

2023



Master of Mathematics (MMath)

University of Exeter

Upper Second Class (2:1) with Honours

📍 Exeter, UK

My master's studies focused on the intersection of mathematics and computer science.

Mathematical Modeling with MATLAB:

- Applied mathematical concepts to real-world problems using MATLAB.
- Developed skills in numerical analysis, optimization, and simulation.

Data Analysis and Statistical Modeling with R and Python:

- Utilized RStudio for data manipulation, visualization, and statistical inference.
- Analyzed complex datasets to derive actionable insights.

Technical Expertise and Applications

Data Analysis

Python: Analyzed a dataset containing approximately 2 million nested data points related to job listings on Indeed (UK) during the COVID-19 pandemic. Calculated key metrics to assess the broader job market, drawing conclusions about the impact of COVID restrictions and their influence on employment trends.

R: Investigated the effects of marine heatwaves on phytoplankton using large spatial-temporal datasets recorded by satellites (e.g., sea-surface temperature and chlorophyll concentration). Employed regression analysis and developed an ARIMA model to describe the causal relationship between these variables.

Modeling and Simulation

MATLAB: Conducted extensive research on option pricing theory for a master's thesis, exploring the binomial option pricing model and the Black-Scholes method. Developed an alternative numerical model for path-dependent (Asian) options, which eliminated the need for path searching, significantly reducing computational complexity while maintaining accuracy. Validated the model through Monte Carlo simulations and corroborated the results with established findings in the literature.

Software Development & Automation

Next.js: Developed a company website using React, Tailwind CSS, and HTML within the Next.js framework. Integrated dynamic features that interact with the company's reporting software API, displaying real-time data on job completions and geographical coverage.

Automation: Automated a manual data entry task by developing a Python script utilizing the Gmail API to scan business emails for new work orders. Extracted information from attached PDFs and populated reporting software autonomously. Deployed the script on a headless Raspberry Pi with remote update capabilities via Git.

CERTIFICATES



- CS50: Introduction to Computer Science (EDX, 2020)
- CS50's Web Programming with Python and JavaScript (EDX, 2021)

PROJECTS

Selenium Chess Bot

Python-based project designed to interface with Lichess. It includes functionality to detect where pieces are on the board using html, calculate the best move, inject JavaScript to display the best move. Automatic releases with the use of github actions.

View on Github -    

Excel to Calendar Web App

Full-stack app to convert Excel shift rosters into calendar events, featuring Excel parsing, Gmail OAuth, and direct Google Calendar integration. Built with Next.js (Vercel) and FastAPI (Render).

View on Github -    

Pricing Asian Options in Matlab

Implemented a pricing model in MATLAB and improved its efficiency by reducing the time complexity of the algorithm.

View on Github -   

MHW and Their Effects on Phytoplankton

Utilized R to analyze large satellite datasets, identifying marine heat waves and their effects on phytoplankton.

View on Github -   

Professional Skills

Python

R

SQL

Web Scraping

API Integration

Data Analysis

Data Visualization

FastAPI / Next.js

Soft Skills and Strengths

Problem Solving

Collaboration

Documentation

Data-Driven Decisions

Stakeholder Communication

Attention to Detail

Project Planning

Time Management

Other Interests

 Gymnastics

 Motorbiking

 Gym

 Climbing

 Travelling

 Chess

Download My CV

Get the latest version of my CV via the QR code below.



WORK EXPERIENCE

May 2025-
Present



Data Scientist Intern
Turn Leaf Analytics

London, UK

Responsibilities & Achievements

- Built Python scrapers to automate retrieval of macroeconomic datasets from dynamic web portals, APIs, and embedded dashboards.
- Programmatically replicated undocumented .NET form interactions to enable reliable data extraction from complex sites.
- Engineered robust pipelines to clean, normalize, and structure time-series data for downstream analytics.
- Evaluated and validated external datasets for reliability, completeness, and analytical suitability, guiding integration into workflows.

Aug 2024-
Present



Sole Trader - Bookkeeping
LDL Contracting Services

Sydney, NSW

Responsibilities & Achievements

- Managed full-cycle bookkeeping, including reconciling accounts and maintaining financial records using Xero and Dext.
- Automated data processing tasks by writing a Python script to convert .txt files to .csv format when the previous bank did not support exporting .csv files.
- Provided detailed financial summaries to assist in decision-making and compliance with tax obligations.

Feb 2023-
Aug 2024



Restoration Technician
RYSA Group LTD

Wagga Wagga, NSW

Responsibilities & Achievements

- Managed data recording and authored detailed restoration reports.
- Developed and deployed a Python-based email pipeline using the Gmail API.
- Designed and launched the company website using the Next.js React framework.

Dec 2013-
Sep 2020

Bookkeeper
East London Pallets LTD

London, UK

Responsibilities & Achievements

- Digitized financial accounts, improving accuracy and efficiency in tracking expenses, sales, and stock.
- Implemented a marginal tax scheme, reducing VAT costs through optimized stock records.
- Created performance dashboards, providing directors with clear, actionable insights.
- Successfully led a change of use application, securing local council approval for the company's strategic growth.
- Developed time-series graphs in R to illustrate the impact of COVID restrictions on business performance, securing government support.

Personal Achievements

- International Competitor:** Represented England at the Cheerleading Worlds competition, placing 5th.
- University Graduate:** Broke new ground as the first in my family to attend university.
- National-Level Gymnast:** Secured 2nd place in the UK at a national gymnastics competition